INCH-POUND

MIL-DTL-13823E (MR)
15 December 2025
SUPERSEDING
MIL-DTL-13823D (MR)
19 November 1998

DETAIL SPECIFICATION

STEEL: SHAPES, FORGED OR FORMED; BARS, FLATS, AND SPECIAL SECTIONS (FOR ARMOR)

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers weldable steel bars, flats and special sections produced from billets to specified dimensions by forging or hot forming, for armored vehicle components.

1.2 Classification. Forged or formed steel armor (FSA) should be of the following classes as specified (see 6.2 and table III).

1.2.1 Class I. Steel shapes, forged or hot formed with as finished hardness range corresponding to class 1 MIL-DTL-12560 Rolled Homogeneous Armor (RHA) steels. For existing engineering drawings that reference MIL-DTL-13823 with no class specified, the default class is class I.

1.2.1.1 Class Ia. Steel shapes, forged or hot formed with as finished hardness range corresponding to class 2 MIL-DTL-12560 RHA steels.

1.2.1.2 Class Ib. Steel shapes, forged or hot formed with as finished hardness range corresponding to class 4a MIL-DTL-12560 RHA steels.

1.2.1.2 Class Ic. Steel shapes, forged or hot formed with as finished hardness range corresponding to class 4b MIL-DTL-12560 RHA steels.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.2.2 Class II. Steel shapes, forged or hot formed with as finished hardness range corresponding to MIL-DTL-46100 High-Hardness Armor (HHA) steels.

1.2.3 Class III. Steel shapes, forged or hot formed with as finished hardness range corresponding to MIL-DTL-46186 Very High Hardness Armor (VHA) steels.

1.2.4 Class IV. Steel shapes, forged or hot formed with as finished hardness range corresponding to MIL-DTL-32332 Ultra High-Hardness Armor (UHA) steels.

1.2.5 Class V. Steel shapes, Steel shapes, forged or hot formed composed of bonded steel layers with properties or functions corresponding to MIL-DTL-46099 Multi-Hardness Armor (MHA) steels.

1.3 Geometries. The steel should be of the following shapes and sizes as specified (see 6.2):

a. Bars, round: 1/4 inch up to and including 4 inch diameter.
b. Bars, square: 1/4 inch up to and including 4 inch.
c. Flats, 1/4 inch or more in thickness: Not over 6 inches wide.
d. Shapes of special sections: In accordance with applicable drawings.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-12560	-	Armor Plate, Steel, Wrought, Homogeneous (For Use in Combat-Vehicles and for Ammunition Testing)
MIL-DTL-32332	-	Armor Plate, Steel, Wrought, Ultra-High-Hardness
MIL-DTL-46099	-	Armor Plate, Steel, Metallurgically Bonded, Dual Hardness (0.187 Inches To 0.700 Inches Inclusive)
MIL-DTL-46100	-	Armor Plate, Steel, Wrought, High-Hardness
MIL-DTL-46186	-	Armor Plate, Steel, Wrought, Very High Hardness

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-662	-	V50 Ballistic Test for Armor
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

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MIL-STD-3040 - Arc Welding of Armor Grade Steel

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://ansi.org/).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A6/A6M - Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes and Sheet Piling

ASTM A751 - Methods, Practices, and Definitions for Chemical Analysis of Steel Products

ASTM D3951 - Practice for Commercial Packaging

ASTM E8/E8M - Standard Test Methods for Tension Testing of Metallic Materials

ASTM El0 - Brinell Hardness of Metallic Materials

ASTM E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

ASTM E23 - Notched Bar Impact Testing of Metallic Materials

ASTM E381 - Standard Method of Macroetch Testing Steel Bars, Billets, Blooms, and Forgings

ASTM E452 - Standard Test Method for Calibration of Refractory Metal Thermocouples Using a Radiation Thermometer

(Copies of these documents are available online at https://www.astm.org).

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Steel shapes shall be in accordance with the applicable drawings, the requirements of this specification, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.4).

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3.2 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required, and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for ballistic acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and ballistic acceptance test samples or sections shall be manufactured by the process proposed for use on production items. The prepared articles for ballistic acceptance shall be submitted to Commander, USA ATC, ATTN: TEDT-AT-SL- V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059. No first article requirements shall be waived without review and approval by the procuring contracting officer (see 3.16 and 6.4).

3.2.1 First time producer. First time producers wishing to qualify to this specification shall follow the instructions of 6.6.

3.3 Chemical composition. A declared chemistry shall be submitted to the contracting agency or its authorized representative. The chemical composition of the declared chemistry shall conform to the requirements of table I, column A, unless otherwise specified in the contract or purchase order (see 6.2). If an analysis of Antimony, Arsenic or Lead is required, it shall be specified in the contract or purchase order (see 6.2). The chemical composition shall be determined by a product analysis (values shall be listed as weight percent) in accordance with 4.3.3, 4.5.1.1, and 4.5.2.1. The first article and the production test samples shall utilize the same declared chemistry within the allowable ranges proposed for use in production. A statement showing the heat analysis of each melt and one product analysis of each lot, complete details of the heat treatment, and all mechanical properties of each lot of production plates shall be furnished for the files of the purchaser at no cost. All elements of the chemical composition specified in table I shall be shown in the statement. Any additives or hardening agents intentionally added shall be declared. The supplier may establish and submit separate limits for each thickness of armor to be furnished.

3.3.1 Carbon (product analysis). The carbon content determined by product analysis by class shall be limited as follows:

class I	0.27 from 0.098 inches to 2.000 inches thick, inclusive 0.29 from 2.001 inches to 4.000 inches inclusive 0.31 from 4.001 inches and greater
class II	0.32
class III	0.35
class IV	0.55
class V	Established by the contractor.

3.3.2 Carbon equivalence. Carbon equivalence (CE) shall be calculated and recorded for each heat per ASTM A6/A6M, i.e.

$$CE = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15}$$

where the elements are expressed in wt%.

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The CE values shall be reported on the completed Formed Steel Armor (FSA) Test Data Form (see figure 1). Carbon equivalence limits by class are as listed in table II.

TABLE I. Chemical composition and precision (product analysis). 4/

ELEMENT	COLUMN A MAXIMUM LIMIT for DECLARED CHEMISTRY (WEIGHT PERCENT)	COLUMN B 5/ ALLOWABLE RANGE for FUTURE PRODUCTION LOTS (WEIGHT PERCENT)
Carbon	See 3.3.1	± 0.05 3/
Manganese	NONE REQUIRED, HOWEVER IF: ≤ 1.00 > 1.00	 ± 0.15 ± 0.20
Phosphorus	0.020 1/	3/
Sulfur	0.010 1/	3/
Silicon	NONE REQUIRED, HOWEVER IF: ≤ 0.60 > 0.60 to ≤ 1.00 > 1.00	 ± 0.10 ± 0.15 ± 0.20
Nickel	NONE REQUIRED 2/	± 0.25
Chromium	NONE REQUIRED, HOWEVER IF: ≤ 1.25 2/ > 1.25	 ± 0.15 ± 0.25
Molybdenum	NONE REQUIRED, HOWEVER IF: ≤ 0.20 3/ > 0.20	 ± 0.035 ± 0.075
Vanadium	NONE REQUIRED 3/	± 0.05
Niobium	NONE REQUIRED 3/	± 0.05
Boron	--- 6/	3/
Copper	0.25 2/	3/
Nitrogen	0.03 2/	3/
Titanium	0.10 2/	3/
Aluminum	0.10 2/	3/
Lead	0.01 2/	3/
Tin	0.02 2/	3/

1/ Phosphorus and sulfur should be controlled to the lowest attainable levels, but in no case shall the combined phosphorus and sulfur contents exceed 0.020 wt%.
2/ When the amount of an element is less than 0.02 percent the analysis may be reported as [< 0.02 wt%].
3/ Product analysis values may not exceed those listed as the maximum limit of the corresponding plate spec.
4/ Elements not listed in table, but intentionally added, shall be reported.
5/ Values are actual tolerance limits NOT percent tolerances. The analysis from the First Article sample is the "Declared Chemistry" which is used to calculate the range.
6/ When the boron is specified in the alloy, its content heat analysis determination shall not exceed 0.003 percent.

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TABLE II. Carbon equivalence limits by class.

Class	Ordered Thickness Ranges (inches)	Carbon (product analysis) (weight percent)	CE (weight percent)
I	< 2.000 2.000 - 4.000 > 4.000	0.27 0.29 0.31	< 0.80 < 0.85 < 0.90
Ia	< 2.000 2.000 - 4.000 > 4.000	0.27 0.29 0.31	< 0.80 < 0.85 < 0.90
Ib	< 2.000 2.000 - 2.750 2.751 - 4.000 > 4.000	0.27 0.29 0.29 0.31	< 0.75 < 0.75 no limit 1/ no limit 1/
Ic	< 2.000 2.000 - 4.000 > 4.000	0.27 0.29 0.31	no limit 1/ no limit 1/ no limit 1/
II	all thicknesses	0.32	< 0.80
III	all thicknesses	0.35	< 0.90
IV	all thicknesses	0.55	no limit 1/
V	all thicknesses	2/	no limits

1/ There is no CE limit however it shall be reported on the Formed Steel Armor (FSA) Test Data Form (see figure 1).

2/ Established by the contractor.

3.4 Heat treatment. The supplier shall select the heat treatment of the material to meet the requirements of this specification. A statement of the heat treatment to be used shall be submitted to the procuring contracting officer. All bars, flats or special sections in each lot, including samples, shall receive the same heat treatment except for such variations in tempering temperature as may be necessary to produce the prescribed hardness. The tempering temperature shall be reported on the Formed Steel Armor (FSA) Test Data Form (see figure 1).

3.5 Heating. Except as provided elsewhere in this specification, local or general heating shall not be performed after final quenching and tempering. A detailed outline of the procedure to be used in each operation of the following processes shall be submitted in writing to the procuring contracting officer for authorization (see 6.2).

3.5.1 Oxygen cutting. Unless otherwise specified (see 6.2), oxygen cutting shall be permitted after final heat treatment provided the procedure is such that no cracks capable of being observed with the unaided eye develop on any oxygen-cut edge.

3.5.2 Repairing. Weld repairs after final heat treatment before or after shipment from the production facility, shall be made only when specifically authorized (see 6.2) and in accordance with MIL-STD-3040 by the procuring contracting officer.

3.5.3 Hot forming. Unless otherwise authorized by the procuring activity, hot forming after the final quenching and tempering operations shall not be performed.

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3.6 Quality. The quality and cleanliness of bars or forging stock shall be in accordance with the relevant sections in ASTM E381.

3.7 Production weldability. The steel bars, flats, or special sections shall be capable of being welded by production welding methods as demonstrated by preparation, for approval by the procuring contracting officer, of samples representing typical production weldments for the intended application in accordance with MIL-STD-3040. This requirement may be waived at the discretion of the procuring contracting officer upon receipt and approval of the production welding characteristics of the type of steel involved.

3.8 Mechanical properties.

3.8.1 Hardness. The average surface hardness of a minimum of four hardness tests per tempering furnace load shall be within the range determined for the ordered class for the applicable thickness. The hardness requirements for each of the classes appear within the hardness requirements sections of the corresponding wrought plate specifications. The diameters of Brinell hardness impressions determined on the surface of any bar, flat, special section or sample shall not vary by more than 0.15 mm respectively between the maximum and minimum values. Cross-sectional hardness tests shall be conducted on impact samples 4 inches or greater in thickness, the average of all hardness tests, both surface and cross-section, shall be within ± 0.05 mm of the range specified for the thickness involved.

3.8.2 Impact resistance. The V-notch Charpy impact resistance shall meet the requirements of the applicable hardness and thickness for the class selected. These requirements are listed in the "Minimum Charpy V-notch impact requirements (at -40°F ± 2°F)" tables that appear within each of the corresponding wrought steel armor plate specifications. There are no impact requirements for class V.

3.8.3 Tensile requirements for special sections. For special sections, the average yield strength and elongation of all specimens shall be within 10% of the average yield strength and 5% of the average elongation unless otherwise specified on the drawing or in the contract or purchase order. The yield strength and elongation readings shall be reported on the Formed Steel Armor (FSA) Test Data Form (see figure 1).

3.8.4 Test blocks for special sections. When the special section dimensions do not permit direct sampling from which to produce mechanical specimens, a test block of sufficient volume shall be prepared following the thermomechanical process of the special section it represents to accommodate the preparation of mechanical test specimens upon approval of the procuring activity.

3.9 Ballistic requirements. Ballistic requirements per critical thickness or thicknesses as established within the contract or purchase order shall be in accordance with the Appendix A of the corresponding RHA, HHA, VHA or UHA specification as applicable. A ballistic test plate must be of the same melt of steel, of the same thickness, having the same treatment, and heat treated using the same thermal cycle process in the same production heating and cooling equipment in the same facility. When a formed part is ordered, a preferred method of meeting this requirement is to use a cutout or drop section of a completed formed part. When specified by the procuring activity (see 6.2), production materials and ballistic test plates shall be allowed to be

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heat treated separately. To facilitate the correct usage of the Appendix A ballistic acceptance requirements from the corresponding wrought plate specifications, cross referencing between the MIL-DTL-13823 classes and their corresponding wrought steel plate classes are provided in table III with additional guidance for the default or preferred classes noted.

TABLE III. MIL-DTL-13823 relationship to wrought steel armor plate specifications.

MIL-DTL-13823 Class	Wrought Plate Specification	Wrought Plate Class
I [1/]	MIL-DTL-12560	1
Ia	MIL-DTL-12560	2
Ib	MIL-DTL-12560	4a
Ic	MIL-DTL-12560	4b
II	MIL-DTL-46100	1
III	MIL-DTL-46186	2 [2/]
IV	MIL-DTL-32332	1 [3/]
V	MIL-DTL-46099	N/A

[1/] Class I is the default when no class for MIL-DTL-13823 is indicated in the engineering drawing, contract or purchase order.

[2/] For MIL-DTL-13823 class III, class 2 VHA Appendix A ballistic acceptance values are used unless otherwise indicated in the contract or purchase order.

[3/] For MIL-DTL-13823 class IV, class 1 UHA Appendix A ballistic acceptance values are used unless otherwise indicated in the contract or purchase order.

3.9.1 Partial penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect: when four (4) partial penetrations are above the minimum requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be specified) above the minimum requirement.

3.9.2 Ballistic test articles greater than 3 inches ordered thickness. Ballistic testing of forged or formed components over 3 inches ordered thickness is only required if requested by the contracting activity. Unless otherwise indicated in the contract or purchase order, ballistic test components greater than 3 inches in thickness shall have a 3 inch diameter lifting hole at the center of the longest edge when possible, with the center hole positioned 3 -0.000 +0.500 inches from the edge of the test article. Additional allowances for mounting (see 4.3.6) may be needed and are authorized to accommodate complex shapes or geometries.

3.9.3 Ballistic testing of forged or formed components. The frequency of ballistically testing production forged or formed components shall be specified in the applicable engineering drawing contract or purchase order (see 6.2).

3.9.4 Ballistic test article information. For each lot, a properly completed Formed Steel Armor (FSA) Test Data Form (see figure 1) shall be submitted with each ballistic test article or section that represents that particular processing lot.

3.10 Dimensions and dimensional tolerances.

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3.10.1 Dimensions. Bars, flats, and special sections shall comply with the dimensions as specified on the applicable drawings or specified in the contract or purchase order.

3.10.2 Tolerances. The thickness of any bars or flats, after final heat treatment, shall not vary by more than the amounts indicted in ASTM A6. Tolerances for special sections shall be as specified in the contract or purchase order.

3.10.2.1 Tolerances for special sections. Tolerance for special sections shall be according to the engineering drawing, contract, or purchase order.

3.11 Identification. Identification markings and records shall be such as to ensure positive identification of all bars, flats and special sections, samples and specimens, with the lot and corresponding heat from which they were produced. The key to identification symbols shall be furnished to the inspector prior to submittal of material for inspection and testing.

3.12 Physical marking for identification. Each bar, flat, and special section or its section or surrogate shall be legibly marked in at least one location in characters not less than 0.5 inch high with the following information.

(a) Manufacturer's name, trademark, or both.
(b) Number of this specification.
(c) Class.
(d) Army part or drawing number.
(e) Lot number.

3.12.1 Ballistic test samples. First article, acceptance, and ballistic test forgings shall be legibly marked with the information in 3.12. In addition, first article bars, flats, and special sections shall be marked "PRE" and acceptance test bars, flats, and special sections marked "ACC". This marking shall be marked in at least two opposing locations in letters not less than 0.5 inch high unless otherwise directed by the engineering drawing, contract, or purchase order.

3.13 Information required. In addition to the information required in the Formed Steel Armor (FSA) test data form (see figure 1), a statement showing the heat analysis of each heat and complete details of the heat treatment of each lot shall be furnished for the files of the procuring activity. All elements of the chemical composition shall be shown in the statement, including special additives or hardening agents, whether shown in table I or not.

3.14 Information required on ballistic test samples. Unless otherwise specified in the contract or purchase order, a properly completed Formed Steel Armor (FSA) Test Data Form (see figure 1) shall be submitted with each ballistic test shape or section containing the information as specified in MIL-STD-662. Ballistic retest articles shall be marked "R1" and "R2" respectively (see Appendix A rejection and retest procedures from the corresponding wrought plate specifications).

3.15 Workmanship. Bars, plates, and special sections shall be free from injurious imperfections including cracks, pits, porosity, laminations, segregation, and inclusions, for their intended use.

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REQUEST FOR BALLISTIC TEST OF FORMED STEEL ARMOR

SPECIFICATION: MIL-DTL-13283	**REVISION: E**	**AMENDMENT:**
FIRING RECORD:	**DATE:**	**TEST CONTRACT NO:**

	SOURCE PLATE	**FORGE**	**HEAT TREATER**
Name:			
Address:			
POC:			
Phone No:			
Email:			

Test Item Identification: _____	Thickness (max): _____ Thickness (min): _____	Austenizing Temp: _____ Quench Medium: _____
Lot No._____ Casting No._____	FSA Classification CLASS: _____ Carbon Equivalence (CE): _____	Tempering Temp: _____ Forging Repaired (y/n): _____ Remarks: _____

PURPOSE: _____ Acceptance _____ First Article _____ Reduced Testing

SAMPLE: _____ Primary _____ Retest (Firing Record No. of Failed Sample_____)

CHEMISTRY ANALYSIS	C	Mn	P	S	Si	Ni	Cr	Mo
FIRST ARTICLE RESULTS:								
ACCEPTANCE (Pass / Fail)								

V:	Nb:	B:	N:		
Cu:	Ti:	Al:			

MECHANICAL PROPERTIES: CHARPY IMPACT @ -40 Degs. F. **Specimen Size:** : _____

BRINELL HARDNESS: _____ **CHARPY VALUES (FT-LBS) 1:** _____ **2:** _____ **3:** _____

TENSILE TESTS FOR SPECIAL SECTIONS ONLY

YIELD STRENGTH (ksi) 1: _____ **2:** _____ **3:** _____ **4:** _____ (special sections only)

ELONGATION (%) 1: _____ **2:** _____ **3:** _____ **4:** _____ (special sections only)

BALLISTIC TEST RESULTS: SEPARATELY HEAT TREATED TEST PLATE: YES _____ **NO** _____

	Obl. (deg)	**Actual Thickness (in)**	**Required V_{50} (ft/sec)**	**Actual V_{50} (ft/sec)**	**Pass/ Fail**
Test:					
Projectile:					
	WROUGHT PLATE SPEC & CLASS V_{50} USED: _____				

DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT & SUPPLIER OFFICIALS

Lot [met] [failed to meet] the ballistic requirements of specification MIL-DTL-13823

Government Representative:	Date:	Supplier Representative:	Date:

FIGURE 1. Formed Steel Armor (FSA) Test Data Form.

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3.16 Waiver of first article test. Suppliers who have previously produced forged or formed bars, flats, and special sections for armor and have not significantly changed production practices and meeting all the requirements of this specification in the past 37 months, will not be required to perform first article tests.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.1.2 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Lot. Unless otherwise indicated in the contract or purchase order, a lot shall consist of bars, flats, and special sections of the same shape, chemistry, and processing submitted for inspection as a unit, provided that the lot does not exceed 300 forgings or formed parts.

4.3 First article inspection. First article inspection, except as contraindicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4. First article inspection shall consist of the following:

(a) Visual examination (see 3.10, 3.12, 3.14, and 4.3.1).
(b) Dimensional confirmation of near net shape (see 3.10).
(c) Chemical analysis (see 3.3, 4.3.3, and 6.5).
(d) Steel Quality (see 3.6 and 4.3.4).
(e) Production weldability (see 3.7 and 4.3.5).
(f) Mechanical properties (see 3.8 and 4.3.6).
(g) Ballistic testing (see 3.9, 4.3.7, 4.7, and 6.6).

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4.3.1 For visual examination. Unless a sampling plan is specified by the procuring contracting officer, all bars, flats or special sections comprising the lot shall be subject to visual inspection to determine compliance with the identification marking (see 3.12) and workmanship (see 3.15) requirements.

4.3.2 Initial dimensional verification. Bars, flats or special sections selected in accordance with 4.3.1 shall be dimensionally interrogated to ensure the initial dimensions and tolerances achieved are sufficient from which to final machine the ordered component (see 3.10).

4.3.3 For chemical analysis. One sample shall be selected from each heat in accordance with ASTM A751.

4.3.4 For steel quality. Unless otherwise specified (see 6.2), samples for steel quality tests shall be taken representing the top, middle and bottom of the first, middle and last usable ingots of each heat. Samples shall be sufficiently large to prepare test specimens required in 4.5.1.2.

4.3.5 For production weldability. When required, the number and size of samples shall be as specified or approved by the procuring contracting officer and welded in accordance with MIL-STD-3040.

4.3.6 For mechanical properties. Samples for mechanical properties tests of bars, flats, or special sections shall be taken at random with frequency in accordance with the contract or purchase order (recommendations are in 6.2.1). Special sections sampling shall be as required in 4.3.6.1. Mechanical property samples shall be of the same thickness or diameter as the bars, flats, or special sections they represent. The samples shall be either as large as the parts represented or shall be of sufficient size to allow for the number of specimens required in 4.5.1.3 to be taken from them. All specimens shall be taken so that every part of the test specimen will be outside of the heat affected zone of any thermally cut edge.

4.3.6.1 Samples for mechanical properties for special sections. Samples for mechanical properties of special sections including tensile (see 3.8.3) shall be taken to represent the regions with highest and lowest cooling rates.

4.3.7 For ballistic testing. For each configuration that the manufacturer has under contract, three or more of the bars, flats, or special sections shall be submitted for ballistic testing in accordance with 3.9 and Appendix A of the corresponding wrought plate specification (see 1.2 and table III). Prior to submission of the bars, flats, or special sections, the tester and the contractor will discuss, using a drawing of the articles, the ballistic test conditions; namely, the component thickness, impact area and projectile type. The number of articles needed for first article ballistic acceptance testing will be determined. The corresponding wrought plate specification and its accompanying class used for ballistic acceptance testing shall be reported in the space provided on the Formed Steel Armor (FSA) Test Data Form (see figure 1). Modifications, cutting, welding, and bracketing will be determined as needed for test mounting and for lifting and manipulation of thicker components (see 3.9.2).

4.4 Quality conformance (acceptance) inspection. Test for acceptance of a single lot shall consist of testing all requirements specified in section 3. For a series of lots, subsequent verification conformance inspection shall consist of the following:

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(a) Visual examination (3.12, 3.14, and 4.3.1).

(b) Dimensional (see 3.10).

(c) Ballistic testing (see 3.9, 4.3.7, and 4.7).

(d) Preparation for shipment (see 5.1).

4.4.1 Visual. Bars, flats, or special sections selected in accordance with 4.3.1 shall be visually examined for compliance with identification marking (see 3.12, 3.14) and workmanship (see 3.15).

4.4.2 Dimensional. Bars, flats or special sections selected in accordance with 4.3.1 shall be dimensionally examined for compliance with dimensions and tolerances (see 3.10).

4.4.3 Ballistic testing. From each lot, one bar, flat or special section shall be submitted for ballistic testing (see paragraph 4.3.7 for supplementary information).

4.4.4 Preparation for shipment. Examination shall be made to determine compliance with the requirements for packaging (see 5.1).

4.5 Tests.

4.5.1 Test specimens.

4.5.1.1 For chemical analysis. Test specimens for chemical analysis shall be in accordance with ASTM A751 (see 6.5).

4.5.1.2 For steel quality tests. Test specimens for steel quality tests shall be prepared as specified in ASTM E381.

4.5.1.3 For mechanical properties.

4.5.1.3.1 Surface hardness. Each furnace load of shapes, bars, flats or special sections, or the corresponding test block for special sections (see 3.8.4) shall be tested for surface hardness in accordance with ASTM E10.

4.5.1.3.2 Impact specimens. At least three Charpy impact test specimens shall be taken from each sample to determine compliance to the requirements of 3.8.2. Specimens from samples less than four inches in thickness shall be taken from the center of the section and at a distance of two times the thickness or diameter from any quenched end as well as outside the heat affected zone of any oxygen cut edge. Specimens from samples four inches or more in thickness shall be taken one inch below the surface and at a distance of two times the thickness or diameter from any quenched end as well as outside the heat affected zone of any oxygen cut edge. Charpy V-notch specimens shall be machined to the dimensions specified in ASTM E23, except that for material having a thickness from 7/32 to 7/16 inches inclusive, the width of the specimens shall be 5 mm (0.197 inch). The notch shall be perpendicular to the surface of the material.

4.5.1.3.3 Tensile specimens for special sections. For special sections, at least four tensile specimens shall be taken from each sample. Two specimens shall each be removed from each special section in the highest and lowest cooling rate regions. Actual location shall be as specified on the drawing or in the contract or order. All specimens shall be in the same orientation with

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respect to maximum metal flow. Specimen geometries shall be the largest dimensions possible while in compliance with ASTM E8. If the special section size is too small for multiple tensile specimens, additional articles shall be furnished to meet the total specimen quantity.

4.6.2 Test methods.

4.6.2.1 Chemical analysis. Chemical analysis shall be conducted in accordance with the applicable method specified in ASTM A751, (see 6.5). The analysis shall comply with the declared composition established in accordance with the requirements of table I (see 3.3).

4.6.2.2 Steel quality tests. Steel quality tests shall be in accordance with ASTM E381.

4.6.2.3 Production weldability. When required, tests of welded samples shall be as specified or approved by the procuring contracting officer.

4.6.2.4 Mechanical properties tests.

4.6.2.4.1 Brinell hardness tests. Brinell hardness tests shall be conducted in accordance with ASTM E10. Hardness values shall conform with 3.8.1.

4.6.2.4.1.1 Surface hardness. All surface scale and decarburization shall be removed from locations where tests are made. Brinell hardness tests shall be conducted by taking a minimum of four hardness tests per furnace load.

4.6.2.4.1.2 Cross-sectional hardness. For samples 7/16 inches or greater in thickness, cross-sectional hardness shall be determined by taking a minimum of four Brinell hardness readings at a location midway between the top and bottom surfaces of the sample and at least two times the thickness or diameter from any quenched end.

4.6.2.4.2 Impact tests. Charpy V-notch impact tests shall be conducted in accordance with ASTM E23 to determine compliance to 3.8.2 requirements. The impact tests shall be made with the specimens at a temperature of -40 ± 2°F. The average impact value obtained from the tests of the three specimens shall be equal to or greater than the value shown in the Minimum Charpy V-notch impact requirements (at -40°F ± 2°F) in the corresponding wrought steel armor plate specification for the hardness as determined in 4.5.2.4.1.1 for material less than 7/16 inch thickness or 4.5.2.4.1.2 for material 7/16 inch or greater in thickness.

4.6.2.4.2.1 Defective test specimen. In the event that the fracture of a test specimen exhibits a lamination which is revealed as an open slit, or sharply defined shelves extending the full width of the fractured surface, the test specimen shall be discarded. Replacement of such specimens shall be in accordance with ASTM A751.

4.6.2.4.2.2 Certification of impact machines. Certification of impact machines shall be in accordance with ASTM E23.

4.6.2.4.3 Tensile tests for special sections. First article tensile tests for special sections shall be conducted in accordance with ASTM E8.

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4.7 Ballistic testing of production forged or formed steel shapes. When specified on the drawing or contract (see 6.2), ballistic testing of production forged or formed steel shapes shall be conducted by the government at a government test facility specified in the contract on order. Testing shall be conducted in accordance with the requirements of Appendix A of the wrought plate specification corresponding to the ordered class cited as listed in table III unless otherwise indicated in the contract or purchase order.

4.8 Rejection and retests.

4.8.1 Rejection. Where one or more test specimens fail to meet the requirements of this specification the lot represented by the specimen or specimens shall be subject to rejection.

4.8.2 Resubmitting new first article test sample. To obtain first article approval, the supplier shall submit a new first article test sample for each sample which fails to meet the requirements of this specification.

4.8.3 Retests for impact resistance. Acceptance retests will be permissible in accordance with 4.8.3.l or, at the option of the manufacturer, individual bars, flats or special sections of a lot may be submitted for acceptance. When this latter practice is followed, all bars, flats or special sections represented by samples yielding acceptable test results shall be considered acceptable. For batch type furnaces, a sample shall be considered to represent a bars, flats or special sections if it is heat treated with the bars, flats or special sections, or if both received the same heat treatment, except that bars, flats or special sections heat treated with the sample which do not meet the requirements shall not be included. For continuous type furnaces, rejectable sample shall be considered to represent all bars, flats or special sections heat treated subsequent to the last acceptable sample and prior to the following acceptable sample.

4.8.3.1 Retests of bars, flats or special sections with low impact resistance after reheat treatment. At the option of the contractor, bars, flats or special sections showing low impact resistance may be retested after reheat treatment in accordance with the requirements for original testing. Bars, flats or special sections which are retempered only to meet hardness requirements will not be subjected to retest for impact resistance.

4.8.4 Chemical analysis retest and rejection. Chemistry retest and rejection shall be in accordance with ASTM A751.

4.8.5 Ballistic retests of production bars, flats and special sections. When a production bar, flat or special section fails to meet the ballistic test requirements of the corresponding wrought plate Appendix A, the bar, flat or special section involved shall be subject to rejection. Failure of any bar, flat or special section to meet the ballistic test requirements of the relevant Appendix A to the ordered class for this specification shall require the selection of the next bar, flat or special section manufactured following notification of failure under the ballistic test.

4.8.5.1 Shipping special sections from the producing facility. No special sections of the die(s) involved shall be shipped from the producing facility until the first otherwise acceptable special section has passed the ballistic tests (see 3.9).

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4.8.5.2 Ballistic retesting of a lot represented by the bars, flats or special sections exhibiting ballistic performance below the requirements of the corresponding Appendix A. At the option of the contractor, a lot represented by the bars, flats or special sections exhibiting ballistic performance below the requirements of the aligning Appendix A, may be retested ballistically after reheat treatment.

4.9 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced, and providing the manufacturer agrees in writing to maintain the same manufacturing procedures. The manufacturer when eligible for reduced or audit testing may at their option, continue to submit one ballistic test sample or section per lot of material. The test facility shall only test the non-required test samples or sections upon manufacturer's loss of reduced testing eligibility.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The steel covered by this specification is military unique. The bars, flats and special sections are intended for use as components on armored combat vehicles, emplacements and the like for protection against armor piercing projectiles, bursting shell and fragments of high explosive ammunition. Fabrication by welding will normally be involved.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specify steel shape, size, and class (see 1.2).
c. When first article sample is required (see 3.2).
d. If the declared chemistry composition can be different (see 3.3).
e. Heat treatment to be used if required (see 3.4).
f. Permission for oxygen cutting (see 3.5.1).
g. Where samples for steel quality tests should be taken if other than in 4.3.4.
h. If ballistic testing of production components over 3" is required (see 3.9.2).

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i. If ballistic forged or formed shapes are authorized to be heat treated separately (see 3.9).
j. If weld repair after final heat treatment is authorized (see 3.5.2).
k. If alternate sampling is indicated (see 6.2.1.1).
l. Packaging requirements (see 5.1).

6.2.1 Frequency of sampling. In the absence of a sampling plan defined in the contract or purchase order, table IV includes the default frequency of sampling.

TABLE IV. Default frequency of sampling.

Length as heat-treated	No. of samples
Greater than 12 feet	1 from each 25 items or 20,000 lbs 1/
Greater than 2 to 12 feet incl. or 100 lbs. or less	1 from each 50 items or 10,000 lbs 1/
2 feet or less or 10 lbs. or less	1 from each 100 items or 5,000 lbs 1/

1/ Whichever is preferred.

6.2.1.1 Alternate sampling. When alternate sampling is indicated, or requested, the procuring activity should follow MIL-STD-1916 and ANSI/ASQ Z1.4 guidance.

6.3 First article. When first article inspection is required, the procuring contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first (how many) production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The procuring contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Waivers or deviations. The procuring contracting officer should coordinate all requests for waiver or deviation to this specification (see 3.16 and 4.9) with the appropriate design and product assurance activities.

6.5 Chemical analysis. Suggested ASTM instrumental methods that can be used for chemical analysis are E415, E282, E484 and E322. ASTM A751 should be consulted for a list of available methods.

6.6 Potential suppliers. Potential suppliers who have not previously supplied forged or formed armor shapes to this specification and wish to have their material ballistically tested, may do so at their own expense. It is recommended that inquiries for such testing be directed to Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

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The armor production process should be inspected by a cognizant government representative to ensure compliance with the details of this specification.

6.7 Alignment of MIL-DTL-13823 classes to wrought steel plate specifications. To ensure the correct requirements are used for any ordered class of MIL-DTL-13823 forged or formed armor shapes, designers, producers, suppliers, and procurement officials should carefully verify with 1.2 and table III to ensure the full and correct requirements including Appendix A ballistic acceptance data from the wrought steel plate specifications for ballistic acceptance are followed in preparation for the contract or purchase order. The wrought steel armor plate specifications contain multiple classes and additional attention to detail should be made to ensure the correct class is indicated and it is documented in Formed Steel Armor (FSA) Test Data Form (see figure 1) or as otherwise directed in the contract or purchase order.

6.8 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors listed in table V.

TABLE V. Conversion factors. 1/

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

1/ Conversion factors can be associated with ASTM E380 entitled “Metric Practice Guide.”

6.9 Subject term (key word) listing.

Ballistic	Brinell hardness	Charpy
Billets	Combat vehicles	Minimum Acceptance

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY - MR

(Project FORG-2025-002)

Review activities:
Army - AR, AT, AV, TE

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